

Task 1: Encrypt mynotes.txt

Steps

1. Create a file named:
 - mynotes.txt
2. Put some sample text inside it.
3. Use a free encryption tool such as Cryptomator or VeraCrypt.
4. Create an encrypted vault/container.
5. Place mynotes.txt inside the encrypted location.
6. Upload the encrypted file/container to Google Drive.
7. Verify that the uploaded cloud copy does not expose the plaintext contents.

Encryption protects data even if an unauthorized person gains access to the stored file. The encryption tool ensures that the plaintext content cannot easily be read without the required encryption credentials.

Task 2: Automatic Assignment Backup

Objective

Create an automatic cloud backup of your college assignments.

Steps

1. Create a folder:
 - College_Assignments
2. Put several assignment files inside it.
3. Install the Google Drive desktop application or use another supported cloud backup/synchronization method.
4. Configure the assignment folder for synchronization/backup.
5. Add or modify an assignment file.
6. Wait for synchronization to complete.
7. Open Google Drive.
8. Verify that the file exists in the cloud.
9. Verification

Check that at least one assignment is available in the cloud copy.

Task 3: CloudWatch Dashboard

Objective

Monitor EC2 CPU utilization using Amazon CloudWatch.

Steps

1. Log in to AWS.
2. Open CloudWatch.
3. Select Dashboards.
4. Select Create dashboard.
5. Give it a name, for example:
 - EC2-Monitoring
6. Add a widget.
7. Select a metric-based widget such as a line graph.
8. Select:
EC2 → Per-Instance Metrics
9. Select:
 - CPUUtilization
10. Select your EC2 instance.
11. Add the widget.
12. Save the dashboard.

Expected Result

- The dashboard displays the CPU utilization of your selected EC2 instance.

Task 4: CloudWatch CPU Alarm

Objective

Generate an email notification when CPU utilization exceeds 60%.

Steps

1. Open CloudWatch.
2. Go to Alarms.

3. Select Create alarm.
4. Select your EC2 instance.
5. Select:
 - CPUUtilization
6. Configure the condition:
 - Threshold: 60%
 - Condition: Greater than 60%
7. Configure an SNS notification topic.
8. Enter your email address.
9. Create the alarm.
10. Check your email.
11. Confirm the SNS subscription if AWS asks for confirmation.

Expected Result

CloudWatch monitors CPU utilization and changes the alarm state when the configured threshold is breached.

Task 5: CloudWatch Logs

Objective

Create a CloudWatch log group and add a sample log event.

Steps

1. Open CloudWatch.
2. Select Log groups.
3. Select Create log group.
4. Name it:
 - /security-lab
5. Create the log group.
6. Create or select an appropriate log stream.
7. Add a sample log event through an appropriate AWS logging mechanism/API/CLI.
8. Example event:
 - 2026-10-05 10:30:00 - Failed login attempt for user admin

9. Open the log stream.
10. Verify that the event appears.